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2023190 FOB

1 Assignment 1

1. As e always has a dimensionless power

$kx \Rightarrow [M^0 L^0 T^0]$

$k \Rightarrow [M^{-x_1} L^{-x_2} T^{-x_3}]$ Assume $x \Rightarrow [M^{x_1} L^{x_2} T^{x_3}]$

For two physical values to be added they must have the same dimension

dimension of A = dimension of B

and the should be equal to dimensions of y.

Assume $y \Rightarrow [M^{y_1} L^{y_2} T^{y_3}]$

$a \Rightarrow [M^{y_1} L^{y_2} T^{y_3}]$

$b \Rightarrow [M^{y_1} L^{y_2} T^{y_3}]$

2. Water will rise faster in the trunk of tree B because of greater contact with walls of the tissues leading to adhesion because of hydrogen bonding

Forces contributing to rise of water,

- Pressure difference - Transpiration creates a pressure difference between the lower and upper parts of the plant.
- Adhesive forces - Water and plant cells
- Cohesive forces - Water and Water

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3. One possible explanation is that the drugs dexedrine and benzedrine are enantiomers of one another and the drug dexedrine is more easily able to act upon its target protein because of its structure.

4. a) To begin with the insecticide acted/effected on all beetles equally until a genetic mutation emerged which made individuals with that mutation resistant to the insecticide. Individuals with the mutation had a higher chance of surviving and thus reproducing passing on the mutation. As a result, new individuals of the species (the newer generations) had inherited the mutation which made them resistant

b) ~~Placebo/ /group~~ control group

5 a) No the graph is not linear as its slope is not constant throughout the x axis.

b) From 88 - 96 the concentration increased by 10 ppm.

c) $Avg = \frac{385 - 339}{2008 - 1980} = \frac{46}{28} = 1.64$ ppm